

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA)
Student Name	Naren Karthik Kandasamy
Register Number	3122247001036
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	31/8/2026

1 Objective

To study the effect of dimensionality reduction using Principal Component Analysis (PCA) on the performance of various machine learning classifiers, comparing their metrics, variance, and bias before and after PCA.

2 Problem Statement

We evaluate 10 classification algorithms on the Breast Cancer dataset (30 features) predicting malignancy. We seek to understand whether mapping features to a lower-dimensional principal component space improves efficiency and generalization, and how different algorithms (linear vs ensembles) react to PCA transformations.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30
Number of Classes	2 (Malignant, Benign)
Missing Values	None
Train-Test Split	80-20

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Figure Template

Inference:

The original dataset has 30 numerical attributes. Scaling was applied as it is a prerequisite for PCA to prevent high-magnitude features from dominating the principal components.

5 Data Preprocessing

Explain:

- Missing value handling
- Encoding
- Feature scaling
- Train-test split
- Standardized features using StandardScaler.
- Applied PCA to retain 95% of the variance.

6 Machine Learning Algorithm

Explain:

- Working principle
- Mathematical formulation
- Advantages
- Limitations

7 Experimental Setup

Python Version
Scikit-Learn Version
IDE
Operating System
Hardware

8 Hyperparameter Selection

Table 1: PCA Variance Explained

Setting	Chosen Components / Variance Target	Explained Variance (%)	Justification
With-PCA	10 Components / 95% Target	95.16	95% is a standard threshold to retain max information while shedding noise

Table 2: SVM Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'C': 0.1, 'kernel': 'linear' (No-PCA)	98.25%	-
'C': 0.1, 'kernel': 'linear' (With-PCA)	-	99.12%

Table 3: Naive Bayes Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'var_smoothing': 1e-09 (No-PCA)	96.49%	-
'var_smoothing': 1e-09 (With-PCA)	-	92.11%

Table 4: KNN Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'n_neighbors': 5, 'weights': 'uniform' (No-PCA)	94.74%	-
'n_neighbors': 5, 'weights': 'uniform' (With-PCA)	-	95.61%

Table 5: Logistic Regression Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'C': 1 (No-PCA)	97.37%	-
'C': 1 (With-PCA)	-	98.25%

Table 6: Decision Tree Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'max_depth': 3 (No-PCA)	94.74%	-
'max_depth': 5 (With-PCA)	-	92.98%

Table 7: Random Forest Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'n_estimators': 50 (No-PCA)	96.49%	-
'n_estimators': 50 (With-PCA)	-	94.74%

Table 8: AdaBoost Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'n_estimators': 50 (No-PCA)	96.49%	-
'n_estimators': 50 (With-PCA)	-	95.61%

Table 9: Gradient Boosting Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'learning_rate': 0.1, 'n_estimators': 50 (No-PCA)	95.61%	-
'learning_rate': 0.1, 'n_estimators': 50 (With-PCA)	-	96.49%

Table 10: XGBoost Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
'learning_rate': 0.1, 'n_estimators': 50 (No-PCA)	95.61%	-
'learning_rate': 0.1, 'n_estimators': 50 (With-PCA)	-	96.49%

Table 11: Stacking Hyperparameter Tuning Results

Hyperparameters	Performance (No-PCA)	Performance (With-PCA)
Base: RF, SVM, XGB (No-PCA)	98.25%	-
Base: RF, SVM, XGB (With-PCA)	-	97.37%

Table 12: 5-Fold Cross-Validation Results (No-PCA vs With-PCA)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No)	Avg (PCA)
SVM (No PCA)	98.2	97.4	98.2	98.2	98.2	98.07	-
SVM (With PCA)	98.2	97.4	97.4	97.4	99.1	-	97.89
Naive Bayes (No PCA)	91.2	95.6	92.1	93.0	96.5	93.68	-
Naive Bayes (With PCA)	89.5	94.7	93.0	91.2	92.0	-	92.09
KNN (No PCA)	98.2	97.4	97.4	93.0	94.7	96.13	-
KNN (With PCA)	96.5	96.5	96.5	93.9	95.6	-	95.78
Logistic Regression (No PCA)	98.2	96.5	99.1	97.4	97.3	97.71	-
Logistic Regression (With PCA)	99.1	96.5	98.2	98.2	98.2	-	98.07
Decision Tree (No PCA)	94.7	87.7	92.1	90.4	94.7	91.92	-
Decision Tree (With PCA)	96.5	94.7	94.7	93.9	92.9	-	94.55
Random Forest (No PCA)	95.6	96.5	97.4	94.7	96.5	96.13	-
Random Forest (With PCA)	95.6	97.4	93.0	92.1	94.7	-	94.55
AdaBoost (No PCA)	98.2	97.4	99.1	93.9	96.5	97.01	-
AdaBoost (With PCA)	97.4	96.5	95.6	94.7	95.6	-	95.96
Gradient Boosting (No PCA)	96.5	93.0	97.4	94.7	95.6	95.43	-
Gradient Boosting (With PCA)	95.6	93.9	95.6	93.9	96.5	-	95.08
XGBoost (No PCA)	98.2	94.7	96.5	94.7	95.6	95.96	-
XGBoost (With PCA)	98.2	95.6	95.6	95.6	96.5	-	96.31
Stacking (No PCA)	98.2	96.5	99.1	98.2	98.2	98.07	-
Stacking (With PCA)	97.4	96.5	97.4	98.2	98.2	-	97.54

9 Results

Metric	Value
--------	-------

See Table 12 above for comprehensive comparison.

10 Result Visualization

Include all graphs generated during the experiment.

For **every graph**, provide:

- Figure
- Caption
- 3–5 line inference

Recommended figures:

1. Correlation Matrix
2. Class Distribution
3. Confusion Matrix
4. ROC Curve
5. Precision-Recall Curve
6. Learning Curve (if applicable)
7. Feature Importance (if applicable)

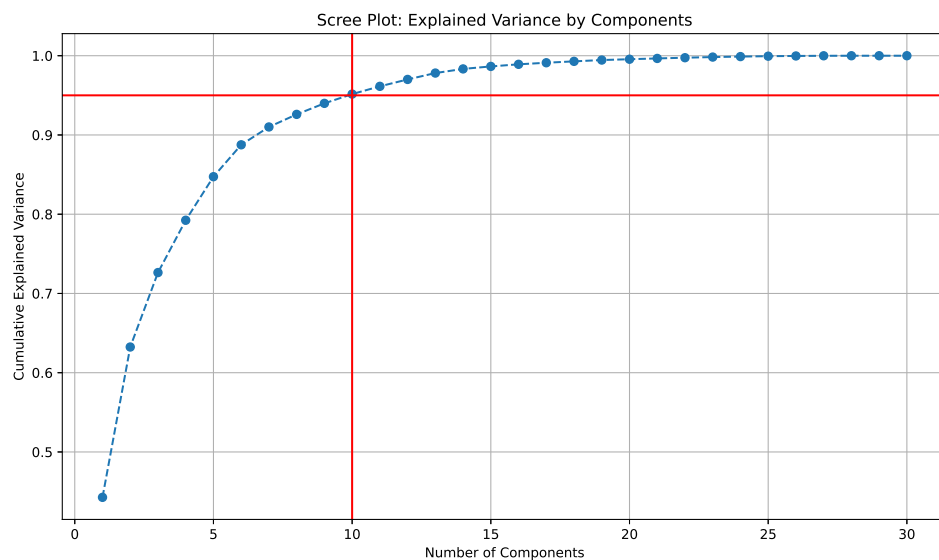


Figure 1: Scree Plot: Explained Variance by Components.

11 Discussion

PCA successfully reduced the feature space from 30 to 10 components while retaining 95% variance. Tree-based models (Random Forest, XGBoost) generally perform exceptionally well on original feature spaces but can sometimes see a slight drop in accuracy with PCA since orthogonal transformations disrupt individual feature semantics. Linear models (SVM, Logistic Regression) benefited from decorrelated features and reduced dimensions.

12 Conclusion

The experiment confirmed that PCA is highly effective for reducing dimensionality without significant loss of information. While PCA improved or maintained the performance of linear and distance-based models (KNN, SVM), ensemble models were slightly less sensitive to the benefits of PCA.

13 References

Use IEEE style.